

Certificate Validator

Exhaustive File-by-File Technical Reference & API Contract Specification

Root / System Configuration

File Name & Path	Role / Title	Detailed Description & Key Logic
package.json	Root Monorepo Manifest	Defines workspaces (`backend`, `frontend`), root utility scripts, and dependency synchronization.
README.md	System Documentation	Comprehensive installation guide, system architecture breakdown, API endpoints list, and security specification.

Backend Entry & Configuration (backend/ & backend/config/)

File Name & Path	Role / Title	Detailed Description & Key Logic
backend/server.js	Express Server Entry	Initializes Express app, configures CORS, JSON body parsers, static file serving for `/uploads`, and registers API route modules.
backend/config/db.js	SQLite Database Setup	Initializes SQLite connection using `better-sqlite3`. Executes table creation DDL statements for `users`, `universities`, `certificates`, `revoked_certificates`, `blockchain_anchors`, `disclosure_proofs`, and `verification_events`.

Backend Controllers (backend/controllers/)

File Name & Path	Role / Title	Detailed Description & Key Logic
authController.js	Authentication Handler	Manages user/university registration, password hashing (bcrypt), login JWT generation, profile fetching, and role authorization.
certificateController.js	Issuance & Revocation	Handles single certificate issuance, Excel bulk issuance (<code>`bulkUploadCertificates`</code>), category normalization, 3-way credential verification, RSA signing, and cryptographic revocation (<code>`revokeCertificate`</code>).
verificationController.js	Verification Engine	Processes online verification requests (<code>`verifyCertificate`</code>), checks replay protection nonces/timestamps, verifies RSA-2048 signatures, checks revocation status, anchors blockchain proofs, and logs verification events.
disclosureController.js	Selective Disclosure	Generates and verifies zero-knowledge predicate proofs (<code>`createPredicateProof`</code> , <code>`verifyPredicateProof`</code>) allowing students to prove criteria (e.g. CGPA >= 8.0) without revealing underlying data.
passportController.js	Skill Passport Engine	Generates shareable public skill passport URLs (<code>`getPublicSkillPassport`</code>), aggregates student credential portfolios, and constructs verified skill badges.
auditController.js	Immutable Audit Feed	Queries system audit logs (<code>`getAuditLogs`</code>), filtered by module, user ID, status, or date range for regulatory compliance.
analyticsController.js	Analytics Aggregator	Computes metrics (<code>`getUniversityAnalytics`</code> , <code>`getStudentAnalytics`</code> , <code>`getVerificationAnalytics`</code>) including issuance volume, monthly trends, and verification pass/fail rates.
chatController.js	AI Chatbot Assistant	Connects to Google Gemini API (<code>`askAIChatbot`</code>) to answer user queries about certificate verification, validation rules, and platform usage.

Backend Middleware & Models (backend/middleware/ & backend/models/)

File Name & Path	Role / Title	Detailed Description & Key Logic
authMiddleware.js	JWT Auth Guard	Intercepts HTTP requests, validates <code>`Authorization: Bearer `</code> , decodes user identity, and enforces role authorization.
userModel.js	User Data Model	Parameterized SQL model for querying, creating, and updating user credentials in SQLite <code>`users`</code> table.
certificateModel.js	Certificate Data Model	Data access methods for inserting certificates, querying by ID/regNo, fetching university certificate lists, and marking revocation status.
verificationModel.js	Verification Event Model	Data access methods for inserting verification activity events and calculating monthly verification counts per university.
walletModel.js	Skill Wallet Data Model	Model for querying student certificate collections, portfolio links, learning goals, and achievement milestones.

Backend Routes (backend/routes/)

File Name & Path	Role / Title	Detailed Description & Key Logic
authRoutes.js	Auth API Routes	Express routes for <code>/api/auth/register`</code> , <code>/api/auth/login`</code> , and <code>/api/auth/me`</code> .
certificateRoutes.js	Certificate API Routes	Routes for <code>/api/certificates/upload`</code> , <code>/api/certificates/bulk`</code> , <code>/api/certificates/revoke`</code> , and university/student certificate fetching.
verificationRoutes.js	Verifier API Routes	Routes for <code>/api/verify`</code> , <code>/api/verifications`</code> , and <code>/api/public-key/:issuer_id`</code> .
disclosureRoutes.js	Selective Disclosure Routes	Routes for <code>/api/disclosure/proof`</code> and <code>/api/disclosure/verify`</code> .
passportRoutes.js	Skill Passport Routes	Routes for <code>/api/passport/me`</code> and <code>/api/passport/public/:passport_id`</code> .
auditRoutes.js	Audit Log Routes	Route for <code>/api/audit/logs`</code> with security filtering.
analyticsRoutes.js	Analytics API Routes	Routes for <code>/api/analytics/university`</code> , <code>/api/analytics/student`</code> , and <code>/api/analytics/verification`</code> .
chatRoutes.js	AI Assistant Routes	Route for <code>/api/chat`</code> connecting to Gemini service.

Backend Services & Utils (backend/services/ & backend/utils/)

File Name & Path	Role / Title	Detailed Description & Key Logic
keyService.js	RSA Key Management	Generates and retrieves 2048-bit RSA key pairs (<code>generateKeyPair`</code>) for university issuers.
qrService.js	QR Synthesis Service	Generates high-density PNG QR code images containing JSON payload, hash, signature, and replay session tokens.
pdfService.js	Server PDF Generator	Compiles server-side downloadable PDF certificates using PDFKit or HTML layout.
walletService.js	Wallet Aggregator	Aggregates student certificates into structured digital skill badges and timeline events.
crypto.js	Cryptographic Utility Core	Provides SHA-256 hash generation (<code>generateHash`</code> ), RSA-2048 signature verification ( <code>verifySignature`</code>), and replay attack session token validation (<code>validateReplayProtection`</code>).
blockchain.js	Blockchain Anchor Mock	Simulates Merkle-tree blockchain anchoring (<code>anchorOnBlockchain`</code> , <code>verifyOnBlockchain`</code>) with transaction hashes and block numbers.
auditLogger.js	Audit Logger	Writes structured security event records to SQLite audit table for all sensitive actions.

Backend Tests (backend/tests/)

File Name & Path	Role / Title	Detailed Description & Key Logic
crypto.test.js	Backend Test Suite	13 unit tests covering RSA keygen, SHA-256 hashing, RSA signatures, tampered data detection, replay protection nonces, revocation anchoring, and selective disclosure.

Frontend Application Core (frontend/src/)

File Name & Path	Role / Title	Detailed Description & Key Logic
main.jsx	React DOM Mount	Mounts `App.jsx` into HTML container and registers PWA service worker.
App.jsx	Router & Transition Provider	Configures React Router v7 routes, opacity page transition animations, and header state.
index.css	Global Design System	Defines CSS design tokens, typography, glassmorphism card styles, button variants, and `.error-msg` banners.
vite.config.js	Vite Bundler Config	Vite build settings, React plugin, and Vite PWA plugin configuration for offline caching.

Frontend API Layer (frontend/src/api/)

File Name & Path	Role / Title	Detailed Description & Key Logic
client.js	Axios Base Client	Configures Axios instance with `baseUrl`, authorization header interceptor, and methods for cert issuance, verification, public keys, and revocation.
analytics.js	Analytics API Client	API client methods for fetching university, student, and verification analytics data.
audit.js	Audit API Client	API client methods for fetching audit logs with search parameters.
blockchain.js	Blockchain API Client	API client methods for inspecting blockchain transaction details.
chat.js	AI Chat API Client	API client for sending prompts to backend Gemini AI chatbot endpoint.
disclosure.js	Disclosure API Client	API client for requesting and verifying selective disclosure predicate proofs.
passport.js	Passport API Client	API client for fetching digital skill passport data.
templateApi.js	Template API Client	API client for fetching template customization settings.
wallet.js	Wallet API Client	API client for fetching student wallet certificate items and portfolio links.

Frontend Pages (frontend/src/pages/)

File Name & Path	Role / Title	Detailed Description & Key Logic
Home.jsx	Landing Page	Public hero landing page showcasing system features, live stats, and portal links.
UniversityDashboard.jsx	University Portal	University dashboard for single issuance, Excel bulk issuance with Drag & Drop, issued cert list with real-time search, and PDF download modal.
UniversityLogin.jsx / UniversityRegister.jsx	University Auth Pages	Login and registration forms for university issuer accounts with issuer code configuration.
UniversityAnalytics.jsx	University Analytics	Visual charts and metrics for university issuance performance.
Verifier.jsx	Certificate Verifier Portal	Multi-mode verifier supporting QR camera scan, drag & drop file upload, cert ID search, offline mode, and batch verification.
VerificationAnalytics.jsx	Verifier Analytics	Analytics on verification trends, pass/fail ratios, and geographic verifications.
StudentDashboard.jsx	Student Dashboard	Student portal displaying issued credentials, QR download, and wallet access.
StudentLogin.jsx / StudentRegister.jsx	Student Auth Pages	Login and registration forms for student accounts.
StudentAnalytics.jsx	Student Skill Analytics	Analytics visualization of student skills, course distribution, and verification counts.
WalletDashboard.jsx	Digital Skill Wallet	Interactive student wallet featuring list/grid views, search filter, cert detail pane, PDF export, and selective disclosure modal.
DigitalSkillPassport.jsx / PublicSkillPassport.jsx	Skill Passport Pages	Shareable digital skill passport views displaying verified credentials and QR share link.
PublicDisclosureView.jsx	Public Disclosure View	Public zero-knowledge verification view for verifying specific predicate proofs.
BlockchainExplorer.jsx	Blockchain Explorer	Interactive blockchain transaction explorer for inspecting certificate Merkle proofs and block details.
AuditLog.jsx	Audit Log Viewer	System audit log viewer with search, filtering, and timestamp inspection.
TemplateManager.jsx	Template Manager	Preview page displaying all 9 certificate category templates.
Register.jsx	Role Selection Register	Hub page directing users to University or Student registration.

Frontend Certificate Templates (frontend/src/components/templates/)

File Name & Path	Role / Title	Detailed Description & Key Logic
TemplateSelector.jsx	Template Category Router	Dynamic component router that normalizes category strings and renders the corresponding template with precise dimensions.
CategoryCertificateTemplate.jsx	Template Wrapper	High-resolution container wrapping template rendering for Canvas export and PDF generation.
GraduationTemplate.jsx	Degree / Graduation Template	Portrait parchment layout featuring gold seals, traditional borders, and academic typography.
CourseCompletionTemplate.jsx	Course Completion Template	Landscape navy/teal layout designed for online and academic courses.
InternshipTemplate.jsx	Internship Certificate Template	Modern professional landscape layout for corporate internship completion.
ProjectTemplate.jsx	Project Completion Template	Clean technical layout for project and capstone completion.
MeritTemplate.jsx	Merit Certificate Template	Elegant gold/crimson layout for academic merit recognition.
DistinctionTemplate.jsx	Distinction Certificate Template	High-contrast luxury layout for graduating with Distinction.
AcademicExcellenceTemplate.jsx	Academic Excellence Template	Formal royal blue layout for top academic standing.
BonafideTemplate.jsx	Bonafide Certificate Template	Official institutional layout for student bonafide certification.
ParticipationTemplate.jsx	Participation Certificate Template	Modern clean layout for hackathons, workshops, and event participation.

Frontend Wallet & AI Components (frontend/src/components/)

File Name & Path	Role / Title	Detailed Description & Key Logic
WalletCertCard.jsx	Wallet Item Card	Visual certificate card component in student wallet with status badges and quick actions.
WalletDetailPane.jsx	Wallet Detail Pane	Expanded detail pane showing certificate metadata, verification status, and disclosure controls.
WalletIndexList.jsx	Wallet Index List	Compact tabular view of wallet certificates for quick browsing.
CertDetailModal.jsx	Cert Detail Modal	Modal view showing high-resolution certificate rendering and PDF download button.
AchievementTimeline.jsx	Achievement Timeline	Chronological timeline visualization of student academic milestones.
LearningGoalTracker.jsx	Goal Tracker	Interactive goal tracking widget for ongoing student courses and certifications.
PortfolioLinksCard.jsx	Portfolio Links Card	Widget for managing student GitHub, LinkedIn, and portfolio links.
FloatingAIButton.jsx	Floating AI Trigger	Floating AI button trigger rendered across application pages.
ChatWindow.jsx	AI Chat Window	Interactive chat interface connecting to backend Gemini AI model.
ChatBubble.jsx	Chat Message Bubble	Formatted chat message bubble supporting markdown and code snippets.
SuggestedQuestions.jsx	AI Suggested Prompts	Clickable sample questions for quick AI assistant queries.
TypingIndicator.jsx	AI Typing Indicator	Animated typing indicator shown while awaiting AI response.

Frontend Utilities & Hooks (frontend/src/utils/ & frontend/src/hooks/)

File Name & Path	Role / Title	Detailed Description & Key Logic
certificateCategory.js	Category Source of Truth	Master category definitions, detail requirement sets, restricted categories, and `normalizeCategoryName()` helper.
certificatePdf.js	Browser PDF Generator	Client-side PDF generator compiling high-resolution certificate PDFs using HTML5 Canvas rendering.
excelParser.js	Excel Bulk Parser	Parses uploaded Excel files using SheetJS, enforces mandatory columns, maps aliases, and normalizes categories.
keyCache.js	Public Key Local Cache	IndexedDB / localStorage cache for university public keys enabling offline verification.
offlineCrypto.js	Offline Verification Engine	Browser-side RSA-2048 signature verification and SHA-256 hash checking for offline mode.
qrDecoder.js	QR Image Decoder	Extracts and parses embedded QR code JSON payload from uploaded PDF/Image files.
revocationCache.js	Revocation Local Sync	Caches synced revocation lists in browser storage for offline revocation checks.
walletStore.js	Wallet Local State	Local state manager for student wallet items and preferences.
useAIProvider.js	AI State Hook	Custom hook managing AI chat conversation history and loading state.
useHeaderHeight.js	Header Offset Hook	Custom hook dynamically measuring navigation bar height for layout spacing.
useReveal.js	Scroll Reveal Hook	Custom hook triggering element entrance animations upon scroll.